

HYUNDAI MOTOR COMPANY

Kano Model & Customer-Centric Quality Management

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1. PROBLEM IDENTIFICATION: VOICE OF CUSTOMER TO QUALITY REQUIREMENTS

In a customer-driven automotive business, customer expectations are continuously collected through surveys, reviews, complaints, service interactions, and ownership experiences. However, every customer request cannot automatically be treated as an equally important product or service requirement. Different customers may have different expectations, while two requirements may also conflict when they are considered for practical implementation.

For a Quality Engineer, the challenge is therefore not simply to collect what customers are asking for, but to understand which expectations actually influence customer satisfaction and which requirements should receive higher priority. If every expressed demand is implemented without structured evaluation, unnecessary complexity, additional cost, and implementation difficulties can increase, which may eventually contribute to quality issues, rework, or corrective work.

The objective is to create a systematic bridge between the Voice of Customer and actual quality or product decisions, while maintaining a practical balance between customer satisfaction, quality, feasibility, and implementation effort.

2. CUSTOMER EXPECTATION ANALYSIS: APPLYING THE KANO MODEL

The Kano Model provides a structured way to understand how different customer requirements influence satisfaction. Instead of treating every customer expectation equally, requirements can be evaluated as Must-Be, One-Dimensional, Attractive, Indifferent, or Reverse characteristics.

- **Must-Be:** Basic expectations where the absence of the requirement can create significant dissatisfaction.
- **One-Dimensional:** Requirements that generally increase customer satisfaction as their performance improves.
- **Attractive:** Requirements that may not always be explicitly expected but can create additional satisfaction when provided.
- **Indifferent:** Features that have little meaningful influence on satisfaction regardless of execution.
- **Reverse:** Characteristics that may actually reduce satisfaction for certain customer groups when implemented.

For Hyundai, the Kano analysis would be developed using customer responses from Kano questionnaires together with customer complaints, reviews, surveys, and service feedback. The purpose is to identify the actual satisfaction impact behind an expressed requirement rather than assuming that every customer request should automatically become a product or service priority.

3. CUSTOMER DATA MANAGEMENT: KANBAN-BASED VISUAL CONTROL

Large volumes of customer information can become difficult to manage when questionnaire responses, complaints, reviews, and service feedback are collected through different channels. A Kanban-based visual management approach can therefore be used to organize the movement of customer-quality information through defined stages of analysis.

The collected information can move systematically from data collection and organization to Kano analysis, validation, prioritization, and finally quality or product decision-making. This approach does not use Kanban as a method for classifying Kano categories; instead, it uses the visual-management principle of Kanban to keep the customer-information flow organized, visible, and traceable.

4. REQUIREMENT VALIDATION & EXPERT QUALITY REVIEW

Customer feedback should remain the primary source of the Voice of Customer, but the final interpretation of a requirement should be supported by sufficient data and quality-engineering experience. A requirement considered essential by one customer may not carry the same importance for another customer group.

Therefore, Kano findings can be reviewed by experienced Quality, Service, Product, and other relevant cross-functional teams. Their role is to interpret evidence and determine whether the requirement should genuinely be treated as a Must-Be, Performance, Attractive, Indifferent, or Reverse characteristic.

5. FROM KANO ANALYSIS TO QFD & QUALITY DECISION

Once customer expectations have been classified through the Kano Model, the prioritized requirements can be taken into Quality Function Deployment (QFD) for further translation. The House of Quality provides a structured method for connecting the Voice of Customer with product or service characteristics and helps identify relationships between different requirements.

This stage becomes particularly important when two customer requirements appear attractive individually but create difficulties or conflicts when considered together. Instead of promising every requested feature independently, QFD allows the Quality Engineer to examine how customer requirements interact and what quality characteristics are required to satisfy them.

The final decision can then be supported through cost-benefit and feasibility analysis, considering customer value, quality impact, implementation effort, competing requirements, and overall business practicality. The lowest-cost option should not automatically be considered the best solution; cost is one decision factor within the overall quality evaluation.

6. ILLUSTRATIVE HYUNDAI CUSTOMER-FACING REQUIREMENT AREAS

For this educational analysis, the following customer-facing areas can be considered for Kano evaluation:

Requirement Area	Customer Expectation & Quality Focus
Vehicle Safety & Emergency	Customer expectations related to safety support and emergency response systems.
Vehicle Reliability	Consistent ownership experience with minimum recurring quality concerns.
After-Sales Service Experience	Quality, consistency, and effectiveness of workshop service interactions.
Service Appointment Convenience	Ease and efficiency of scheduling and managing service requirements.
Mobile/Home Service Convenience	Availability of service support without requiring a workshop visit for every issue.
Warranty Support & Clarity	Customer understanding and hassle-free experience of warranty-related support.
Vehicle-Health & Maintenance	Clear information that helps customers understand vehicle condition and requirements.
Remote Vehicle Convenience	Connected functions providing useful remote access and vehicle-status information.
Digital Ownership Experience	Convenience provided through Hyundai's digital and connected customer ecosystem.
Customer Communication	How effectively customer concerns are received, addressed, and closed.

7. INTEGRATED QUALITY MANAGEMENT FRAMEWORK

The proposed framework connects the major quality-engineering concepts in a sequential manner:

Process Flow: Voice of Customer (VoC) → Kanban Visual Control → Kano Model Analysis → Expert Review & Re-validation → QFD (House of Quality) → Feasibility & Cost-Benefit Analysis → Quality Decision & Implementation

Voice of Customer provides the raw customer expectations through questionnaires, complaints, reviews, and service feedback. The Kano Model evaluates how these expectations influence satisfaction, while a Kanban-based visual workflow helps maintain systematic movement and traceability of the collected information.

After Kano analysis, experienced cross-functional review and customer re-validation can be used where necessary. The prioritized requirements can then enter QFD/House of Quality, where customer expectations are translated into quality and product requirements. Finally, cost-benefit and feasibility analysis supports the decision regarding which requirements should become actual quality or product priorities.

The objective is not to reject customer requirements or to implement every request blindly. The objective is to listen carefully, understand objectively, prioritize systematically, and implement responsibly.